

Harshwardhan Patil

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🌐 Portfolio

Overview

Full-Stack Engineer with **3+ years of experience** building scalable, distributed systems and high-performance APIs across frontend, backend, and cloud environments. Skilled in **system design**, CI/CD on AWS, production monitoring, and delivering **fault-tolerant, data-consistent services** at scale.

Technical Skills

- **Languages & Frameworks:** Java, TypeScript, JavaScript, SQL, Spring Boot, Angular, Node.js
- **Cloud & DevOps:** AWS (EC2, ECS, S3, RDS, CloudFront, ALB, Secrets Manager, Lambda), Docker, Jenkins, GitHub Actions, Nginx, Cloudflare, CI/CD
- **Databases:** PostgreSQL, Redis, MySQL, Neo4j, Elasticsearch, MongoDB, Schema design, Database indexing, Query optimization, Caching strategies
- **Testing & Security:** JUnit, Jasmine, SonarQube, Postman, JWT, RBAC, HTTPS/TLS, CORS, CSRF/XSS protection
- **Design & Architecture:** System Design, Design Patterns, SOLID principles, Distributed backend services, Concurrency handling, Session management

Professional Experience

Software Engineer, Clarivate Analytics – Bangalore, India

Aug 2023 – Aug 2024

- Designed scalable Spring Boot REST APIs with DTO contracts and standardized error handling, reducing breaking changes by **30%** across releases.
- Designed distributed backend services handling concurrent user load with controlled multi-session logins and token-based session management.
- Designed and optimized relational schemas and queries for large analytical datasets, using indexing and query tuning to ensure data correctness.
- Containerized and built deployment pipelines using Docker, Jenkins, and Spinnaker, supporting **50+ production releases** with **zero rollbacks**.
- Proactively identified performance bottlenecks in reporting services and optimized API and cache layers, reducing response latency by **25–35%**.
- Implemented role-based access control across backend and frontend layers, enforcing tiered and region-based data visibility for **10+ access variants**.
- Designed and implemented low-latency typeahead and autocomplete services, improving search response times by **40%** over large-scale datasets.
- Led and developed dynamic PPT and PDF export pipelines, generating **100+ customized reports** with user-selected filters, layouts, and metadata.

Associate Software Engineer, Clarivate Analytics – Bangalore, India

Aug 2021 – Jul 2023

- Optimized frontend rendering and state management for data-heavy views, improving responsiveness of interactive charts and tables by **20–25%**.
- Integrated Elasticsearch into Spring Boot to provide fast, advanced queries across **millions of drug records**, reducing search latency by **30%**.
- Improved frontend performance by minimizing unnecessary re-renders and optimizing data-fetching patterns, resulting in **20% faster page loads**.
- Migrated a legacy client-side component to an Angular 13 application, rebuilding **5+ core business flows** with a modular architecture.
- Implemented secure request-handling mechanisms, including JWT authentication, CORS/CSRF protection, and input validation in **multiple services**.
- Built reusable, aggregation-driven Angular chart components using D3.js, powering **15+ analytics views** for interactive reporting services.
- Architected a highly configurable, reusable Angular table framework adopted across **3 internal teams**, reducing duplicated frontend effort by **50%**.

Education

University of Texas at Arlington

Master's in Software Engineering

May 2026

BMS College of Engineering, India

Bachelor's in Mechanical Engineering

July 2021

Projects

TinyURL — URL Shortener 🌐

Feb 2026 – Mar 2026

Production-grade URL shortening service with QR code generation, link expiry, and a fully automated cloud deployment pipeline.

Tools: Spring Boot, Angular, PostgreSQL, AWS (EC2, RDS, S3, CloudFront, ALB), Docker, Cloudflare, Lambda, EventBridge, SNS, SQS DLQ, CloudWatch

- Deployed **distributed architecture** separating SPA (S3 + CloudFront) from REST API (EC2 + ALB) with **Cloudflare** for DNS, SSL, and DDoS protection.
- Hardened the API layer with **nginx-level rate limiting** (per-IP limits on URL creation, redirects, and connections), request body constraints, and security headers (HSTS, X-Frame-Options, CSP), blocking common scanner User-Agents and vulnerability scan paths before requests reach the application.
- Designed serverless infrastructure scheduler — reducing monthly EC2/RDS costs by **52%** with full observability and zero operational overhead.
- Built **REST APIs** for URL shortening with configurable **TTL-based expiry**, short-code generation, and redirect resolution, backed by **PostgreSQL on RDS**.
- Implemented a **CI/CD pipeline** via GitHub Actions to automate production builds, S3 sync, and **CloudFront cache invalidation** on every push to main.

Employee Management System 🌐

Dec 2024 – May 2025

Enterprise-grade system for managing employees, teams, projects, and access control with production-grade deployment and observability.

Tools: Spring Boot, Angular, PostgreSQL, AWS (EC2, RDS, S3, Secrets Manager), Redis, Docker, GitHub Actions, Jenkins

- Designed a multi-tenant backend enforcing **RBAC** across employees, departments, and projects with data isolation and automated AWS deployment.
- Implemented **REST APIs** for multi-entity services (employees, departments, projects, tasks) with pagination, filtering, and search for fast data retrieval.

RingNet — Graph-Based Fraud Ring Detection 🌐

May 2026

Property graph system detecting fraud networks through shared identifiers and multi-hop traversal — where graph databases outperform relational.

Tools: Neo4j, Java, Cypher, Docker, Maven, Graph Data Science Library

- Built a **property graph** across 150 accounts and 430 transactions — fraud ring traversal is constant-complexity in **Cypher** vs exponential SQL joins.
- Documented graph vs relational trade-off in an **ADR** with side-by-side SQL vs Cypher comparison, making the architectural decision concrete.

Battle Arena – Real-Time Multiplayer Platform 🌐

May 2025 – Present

Real-time multiplayer system featuring a modular game engine, physics-based simulation, and distributed backend services.

Tools: Angular, Phaser 3, Spring Boot, Node.js, MongoDB, WebSockets, Docker, AWS

- Built a distributed multiplayer backend architecture, evaluating polling vs WebSockets and custom protocols for latency, consistency, and scale.
- Architected a **modular, terrain-aware game engine** with clear separation between **physics simulation, collision constraints**, and gameplay rules.